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SUPERSEDING
Specification FXS-305
1 September 1942

MILITARY SPECIFICATION

QUADRANT, ELEVATION, M1

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 This specification covers one type of elevation quadrant known as Quadrant, Elevation, M1, used for laying vertical angles for indirect fire.

2. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS, AND DRAWINGS

2.1 Specifications.—The following specification, of the issue in effect on date of invitation for bids, form a part of this specification:

U. S. ARMY SPECIFICATION

51-7—Instruments, Fire Control; General Specification Governing the Manufacture and Inspection of.

(*Army.*—Copies of specifications should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(*Navy.*—Copies of Military specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Commanding Officer, Naval Supply Depot, Scotia 2, N. Y. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Drawings.—The following drawing of the issue in effect on date of invitation for bids, forms a part of this specification:

ARMY ORDNANCE CORPS DRAWING

92-23-1—Quadrant, Elevation, M1.

(Copies of drawings should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 General requirements and associated tests common to all fire control instruments as contained in Specification 51-7 shall be mandatory in so far as they are applicable.

3.2 Materials.—Materials shall be in accordance with drawings, general specifications and material specifications forming part of this specification.

3.3 Accuracy.—With the quadrant mounted as shown on the assembled view drawings, with the mounting surface of the bracket (plane of faces of bolt hole bosses) in a vertical plane, the locating key horizontal, and the level bubbles centered, the following requirements shall be met:

3.3.1 *Elevation.*—The micrometer and elevation scale shall indicate zero within the width of a graduation line on the micrometer scale.

3.3.2 *Crosslevel.*—The crosslevel bubble shall remain centered within one graduation when the mounting bracket, while main-

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tained in a vertical plane, is elevated and depressed throughout the range of the instrument, with the elevation level bubble kept centered by means of the elevation mechanism.

3.3.3 *Error.*—The error in elevation readings shall not exceed ± 0.4 mil for any setting between minus 200 and plus 800 mils and ± 0.6 mil for any setting between 800 and 1200 mils.

3.3.4 *Backlash.*—The backlash in the elevation mechanism shall not exceed 0.2 mil at any setting.

3.3.5 *Stop rings.*—The stop rings shall allow not less than 5 and not more than 25 mils elevation movement below 200 mils depression and above 1200 mils elevation.

3.4 Illumination. — The lighting system shall illuminate the level bubbles, scale, and micrometer so that readings may be readily made in a dark room when a connector supplying 6-volt d.c. is plugged in the socket and the switch is turned on.

3.5 Temperature. — The temperature requirements specified in Specification 51–7 for type III instruments shall apply.

3.6 Workmanship.—Workmanship shall be in accordance with Specification 51–7.

4. SAMPLING, INSPECTION AND TEST PROCEDURE

4.1 Inspection.—Unless otherwise specified each quadrant and spare part shall be inspected for conformance with this specification and as specified in Specification 51–7.

4.2 Tests.

4.2.1 *Accuracy test.*—Each quadrant shall be tested for accuracy of elevation readings at not less than 12 different settings throughout the scale.

4.2.2 *Temperature test.*—The temperature test contained in Specification 51–7 for type III instruments shall apply.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.—Packaging, packing and marking, shall be in accordance with instructions forming a part of the contract or furnished by the contracting officer in connection with the contract.

6. NOTES

6.1 Ordering data.—Invitation for bids and contracts should specify the following:

a. Title, number and date of this specification.
b. Type of packing required. (See 5.1.)

Notice.—This specification, together with specifications and drawings pertaining to it and bearing a "NOTICE" of similar restrictions, is intended for use only in connection with procurement by the United States Government; and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

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Army—Ordnance Corps

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